

Dynamical Lie Algebras and Their Computation

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Introduction to PQC and DLA



Preliminaries: Lie Algebras & Matrix Lie Groups

Setup: $\mathbb{K} = \mathbb{R}, \mathbb{C}$; $\mathfrak{g}$: finite-dimensional vector space over $\mathbb{K}$.

Lie Algebra

$(\mathfrak{g}, [\cdot, \cdot])$, bilinear bracket $[\cdot, \cdot]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$:

- ① Skew-symmetry: $[X, Y] = -[Y, X]$
- ② Jacobi identity:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

Matrix Lie Group

A **matrix Lie group** is a **closed subgroup** of $GL_N(\mathbb{K})$.

Key Examples (Core Focus)

1. General Linear Lie Algebra

$$\mathfrak{gl}_N(\mathbb{K}) = M_N(\mathbb{K}), \quad [A, B] = AB - BA$$

2. Unitary Matrix Lie Groups

$$U(N) = \{U \in M_N(\mathbb{C}) \mid U^\dagger U = I\},$$

$$SU(N) = \{U \in U(N) \mid \det U = 1\}.$$



Preliminaries: Lie Algebra of Matrix Lie Group

Theorem

Let G be a matrix Lie group. The set

$$\mathfrak{g} := \{X \in M_N(\mathbb{K}) \mid \forall t \in \mathbb{R}, e^{tX} \in G\}$$

forms a Lie algebra over $\mathbb{K}$. We denote $\mathfrak{g} = \text{Lie}(G)$.

Example

Take $G = \text{U}(N)$. We have

$$\text{Lie}(\text{U}(N)) = \mathfrak{u}(N) = \{X \in M_N(\mathbb{C}) \mid X^\dagger = -X\}.$$

Proof sketch: Assume $X \in \text{Lie}(\text{U}(N))$, then

$$(e^{tX})^\dagger e^{tX} = I, \quad \forall t \in \mathbb{R}.$$

Differentiate both sides with respect to t :

$$X^\dagger e^{tX^\dagger} e^{tX} + e^{tX^\dagger} X e^{tX} = 0.$$

Set $t = 0$: $X^\dagger + X = 0 \implies X^\dagger = -X$.



Background of Parameterized Quantum Circuits

- Given a Hermitian operator $H \in \text{Herm}(N)$, the Schrödinger equation reads

$$i\frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle.$$

- If H is time-independent, the solution is unitary evolution $U(t) = e^{-itH} \in \text{U}(N)$.
- Consider piecewise constant Hamiltonians $H_1, \dots, H_L$.
- For each segment with fixed H_ℓ , the unitary block reads

$$U_\ell(\theta_\ell) = e^{i\theta_\ell H_\ell} \in \text{U}(N).$$

- Concatenating all unitary blocks yields a **parameterized quantum circuit (PQC)**:

$$U(\boldsymbol{\theta}) = \prod_{\ell=1}^L e^{i\theta_\ell H_\ell}.$$



PQC and Dynamical Lie Algebra

Given Hamiltonian generators $\mathcal{G} = \{H_1, \dots, H_L\} \subset \text{Herm}(N)$.

Parameterized Quantum Circuit

A PQC is defined by the unitary map

$$U(\boldsymbol{\theta}) = \prod_{\ell=1}^L e^{i\theta_{\ell} H_{\ell}}.$$

Lemma

Let $G_{\mathcal{G}} \subset \text{U}(N)$ be the connected Lie subgroup generated by all $U(\boldsymbol{\theta})$. Then

$$\text{Lie}(G_{\mathcal{G}}) = \langle iH_1, \dots, iH_L \rangle_{\mathbb{R}, \text{Lie}} \subset \mathfrak{u}(N).$$

Definition: Dynamical Lie Algebra (DLA)

The dynamical Lie algebra $\mathfrak{g}_{\text{DLA}}$ is defined as

$$\mathfrak{g}_{\text{DLA}} := \langle iH_1, \dots, iH_L \rangle_{\mathbb{R}, \text{Lie}} \subset \mathfrak{u}(N).$$

Variational Loss Function for PQC

Recall the PQC unitary ansatz:

$$U(\boldsymbol{\theta}) = \prod_{\ell=1}^L e^{i\theta_{\ell} H_{\ell}}.$$

Given initial state ρ_0 , the evolved density matrix is

$$\rho(\boldsymbol{\theta}) = U(\boldsymbol{\theta})\rho_0 U(\boldsymbol{\theta})^{\dagger}.$$

Optimization Objective

Given a target observable O , we optimize $\boldsymbol{\theta}$ to make $\rho(\boldsymbol{\theta})$ approximate O .

- Hilbert-Schmidt loss:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2} \|\rho(\boldsymbol{\theta}) - O\|_{\text{HS}}^2 = \frac{1}{2} \text{Tr}(\rho(\boldsymbol{\theta})^{\dagger} \rho(\boldsymbol{\theta})) + \frac{1}{2} \text{Tr}(O^{\dagger} O) - \text{Tr}(\rho(\boldsymbol{\theta}) O).$$

- $\text{Tr}(\rho^{\dagger} \rho)$ and $\text{Tr}(O^{\dagger} O)$ are constant over $\boldsymbol{\theta}$.
- Minimizing $\mathcal{L}(\boldsymbol{\theta})$ is equivalent to maximizing the overlap

$$\ell(\boldsymbol{\theta}, O) = \text{Tr} (U(\boldsymbol{\theta})\rho_0 U(\boldsymbol{\theta})^{\dagger} O).$$



Barren Plateaus: Gradient Variance Bound

Theorem [Hayes et al., 2022]

The variance of the loss function satisfies

$$\text{Var}_{\boldsymbol{\theta}} [\ell(\boldsymbol{\theta}; \rho_0, O)] \lesssim \frac{\text{Tr}[\rho_0^2] \text{Tr}[O^2]}{\dim \mathfrak{g}_{\text{DLA}}}.$$

Theorem [Ragone et al., 2023]

There exists a constant $\alpha > 0$ such that

$$\text{Var}_{\boldsymbol{\theta}} [\ell(\boldsymbol{\theta}; \rho_0, O)] = \frac{\alpha}{\dim \mathfrak{g}_{\text{DLA}}}.$$

Special case: $\mathfrak{g}_{\text{DLA}} = \mathfrak{u}(N)$

- For an n -qubit system, $N = 2^n$, $\dim \mathfrak{u}(N) = N^2 = 4^n$.
- The gradient variance scales as $\text{Var}_{\boldsymbol{\theta}}[\ell] = O(1/4^n)$.

Physical Interpretation: If $\text{Var}[\ell] = O(1/b^n)$ with $b > 1$, barren plateaus emerge exponentially fast as qubit number n increases.



Towards Efficient Computation of DLA



DLA: Definition and Iterative Computation Framework

Given skew-Hermitian PQC generators $\{B_1, \dots, B_k\}$, we aim to compute a full basis of $\mathfrak{g}_{\text{DLA}}$.

Theorem (Iterative Closure Algorithm)

Define nested subspaces recursively:

$$\begin{cases} \mathfrak{g}_0 = \text{span}_{\mathbb{R}}\{B_1, \dots, B_k\} \\ \mathfrak{g}_{m+1} = \mathfrak{g}_m + \text{span}_{\mathbb{R}}\{[A, B] \mid A, B \in \mathfrak{g}_m\} \end{cases}$$

The sequence stabilizes after finitely many steps M :

$$\mathfrak{g}_{M+1} = \mathfrak{g}_M = \mathfrak{g}_{\text{DLA}}.$$

Critical Drawback: Direct matrix-based iteration suffers exponential computational complexity. We propose a symbolic binary algorithm based on Pauli operators to accelerate DLA computation.



Definition of the n -qubit Pauli Group

Notation: Single-qubit Pauli matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Definition (Pauli Group $\mathcal{P}_n$)

For an n -qubit system, the Pauli group $\mathcal{P}_n$ collects all tensor products of single-qubit Pauli matrices multiplied by global phases:

$$\begin{aligned} \mathcal{P}_n &= \{ \alpha P \mid \alpha \in \{\pm 1, \pm i\}, P \in \{I, X, Y, Z\}^{\otimes n} \} \\ &= \left\{ i^k X^{a_1} Z^{b_1} \otimes \cdots \otimes X^{a_n} Z^{b_n} \mid k \in \mathbb{Z}, a_j, b_j \in \{0, 1\} \right\}. \end{aligned}$$

where the tensor-product Pauli strings are defined as

$$\{I, X, Y, Z\}^{\otimes n} = \{P_1 \otimes P_2 \otimes \cdots \otimes P_n \mid P_j \in \{I, X, Y, Z\}\}.$$



Core Algebraic Rules for Pauli Operators

- Let $a = (a_1, \dots, a_n), b = (b_1, \dots, b_n) \in \mathbb{F}_2^n$;
- Binary Encoding $v = (a, b) \in \mathbb{F}_2^n \times \mathbb{F}_2^n$ define phase-free Pauli string

$$P(v) = X^{a_1} Z^{b_1} \otimes X^{a_2} Z^{b_2} \otimes \dots \otimes X^{a_n} Z^{b_n}.$$

- Two standard bilinear forms over $\mathbb{F}_2^{2n}$:

$$\gamma(v, w) = v^T \Gamma w,$$

$$\Gamma = \begin{pmatrix} 0 & 0 \\ I_n & 0 \end{pmatrix},$$

$$\omega(v, w) = \gamma(v, w) + \gamma(w, v) = v^T \Omega w,$$

$$\Omega = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}.$$

Theorem: Multiplication and Commutation Identities

$$i^k P(v) \cdot i^l P(w) = i^{k+l} (-1)^{\gamma(v,w)} P(v+w),$$

$$P(v)P(w) = (-1)^{\omega(v,w)} P(w)P(v).$$

Skew-Hermitian Basis and Commutator Identity

Theorem: Real Basis of $\mathfrak{u}(2^n)$

The set $\mathcal{B} = \{iP \mid P \in \{I, X, Y, Z\}^{\otimes n}\}$ forms an $\mathbb{R}$ -linear basis of $\mathfrak{u}(2^n)$.

Define vector-labeled skew-Hermitian basis element:

$$B(v) = i^{1+\omega(v,v)} P(v).$$

Pauli Commutator Rule

$$[B(v), B(w)] = \begin{cases} \pm 2B(v+w) & \omega(v, w) = 1, \\ 0 & \omega(v, w) = 0. \end{cases}$$

Remark: Matrix commutator calculations reduce to simple binary vector arithmetic over $\mathbb{F}_2^{2n}$.



Vector-Encoded DLA Closure Algorithm (Pauli Generators)

Algorithm 1 Symbolic DLA Closure for Pauli Generators

Input: Generator binary vectors $\{v_1, \dots, v_k\} \subset \mathbb{F}_2^{2n}$

Output: Binary vector basis spanning the closed DLA

1. Map each generator to its binary vector, collect linearly distinct vectors as $\mathcal{V}$
 2. **while** True **do**
 3. Generate candidate vectors: $C = \{v + w \mid v, w \in \mathcal{V}, \omega(v, w) = 1\}$
 4. Merge sets: $T = \mathcal{V} \cup C$
 5. **if** $|\mathcal{V}_{\text{new}}| = |\mathcal{V}|$ **then**
 6. $\mathcal{V} \leftarrow \mathcal{V}_{\text{new}}$; **Break**
 7. **end if**
 8. $\mathcal{V} \leftarrow \mathcal{V}_{\text{new}}$
 9. **end while**
 10. **return** $\mathcal{V}$
-

Advantage: The complexity is reduced from exponential matrix operations to polynomial binary Gaussian elimination with cost $O(n^3)$.



Numerical Experiments and Closed-Form Conjectures

Notation: $X_i := I^{\otimes(i-1)} \otimes X \otimes I^{\otimes(n-i)} = I \otimes \dots \otimes I \otimes X \otimes I \dots \otimes I$.

Three typical generator families for transverse Ising PQC:

- ① **Case 1: Nearest-Neighbor ZZ + All Single-Qubit X** $\mathcal{G} = \{X_i\}_{i=1}^n \cup \{Z_i Z_{i+1}\}_{i=1}^{n-1}$
- ② **Case 2: Full Pairwise ZZ + All Single-Qubit X** $\mathcal{G} = \{X_i\}_{i=1}^n \cup \{Z_i Z_j\}_{1 \leq i < j \leq n}$
- ③ **Case 3: Global n -fold Z + All Single-Qubit X** $\mathcal{G} = \{X_i\}_{i=1}^n \cup \{Z_1 Z_2 \dots Z_n\}$

Table: Conjectured DLA Dimensions for Three Models

Model	Conjecture for $\dim \mathfrak{g}_{\text{DLA}}$
Case 1 (Nearest $ZZ + X$)	$\dim = n(2n - 1), \quad \mathfrak{g} \cong \mathfrak{so}(2n)$
Case 2 (Full pairwise $ZZ + X$)	$\dim = 2^{2n-1} - 2$
Case 3 (Global $Z + X$)	$\dim = 3 \cdot 2^{n-1}$

Numerical scanning up to $n = 10 \sim 20$ qubits motivates these closed-form dimension conjectures.

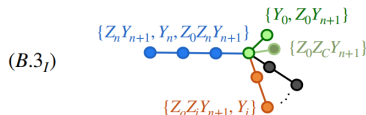
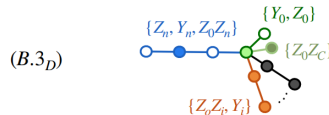
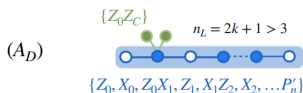
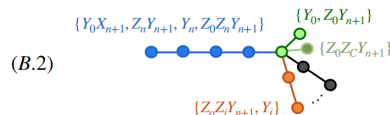
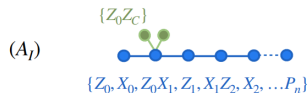


Recent Progress & Open Problems

Pauli DLA Classification Result (Aguilar et al., 2024)

Lie algebras generated by Pauli strings split into two families:

- ① **Polynomial dimension:** free-fermion type ($\mathfrak{so}$, $\mathfrak{sp}$)
- ② **Exponential dimension:** large subalgebras inside $\mathfrak{u}(2^n)$



Theorem (Classification of Pauli Lie Algebras, Aguilar et al., 2024)

Let $\mathcal{G}$ be a set of Lie-algebraically independent Pauli operators, whose anti-commutation graph Γ is connected. The corresponding Pauli Lie algebra decomposes as follows:

- $\bigoplus_{i=1}^{2^{n_c}} \mathfrak{so}(n_L + 1)$, if Γ reduces to $\bar{\Gamma} \in \text{Class } A$;
- $\bigoplus_{i=1}^{2^{n_c}} \mathfrak{sp}(2^{n_2})$, if Γ reduces to $\bar{\Gamma} \in B1$;
- $\bigoplus_{i=1}^{2^{n_c}} \mathfrak{so}(2^{n_2+3})$, if Γ reduces to $\bar{\Gamma} \in B2$;
- $\bigoplus_{i=1}^{2^{n_c}} \mathfrak{su}(2^{n_2+2})$, if Γ reduces to $\bar{\Gamma} \in B3$.



Open Problems in Pauli DLA Theory

- 1 Unified graph-theoretic criterion: characterize which interaction topologies yield polynomial-scaling DLAs.
- 2 Rigorous connection between DLA root structure, circuit expressibility and barren plateau gradients.
- 3 Extend the complete classification framework to k -local ($k \geq 3$) Pauli generating sets.
- 4 Relate infinite-depth DLA closure to finite-depth PQC reachable unitary orbits.
- 5 Efficient criteria to decide classical simulability for general polynomial-size Pauli DLAs.



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All derivations, algorithms and numerical conjectures presented in this work are completed independently by the author.

Thank You!

